VISVESVARAYA TECHNOLOGICAL UNIVERSITY
Jnana Sangama, Belagavi: 590018 Karnataka, India

A Project Report
On

"IOT-CONTROLLED AUTOMATED MULTI-FIELD IRRIGATION SYSTEM"

Submitted in partial fulfilment of the requirements of the award of the degree of

BACHELOR OF ENGINEERING
in
Electronics and communication Engineering

Submitted by

1. ABHILASH N **(4AI22EC002)**

2. AMITH K R **(4AI22EC006)**

3. CHIRAAG A E **(4AI22EC024)**

4. MANASA T P **(4AI22EC057)**

Under the guidance of

Dr. Kumuda T
B.E., MTech., Ph.D., LMISTE., MIE.,
Professor
Dept. of E&CE
AIT, Chikkamagaluru.

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
ADHICHUNCHANAGIRI INSTITUTE OF TECHNOLOGY
CHIKKAMAGALURU, KARNATAKA, INDIA
2025-2026

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belagavi: 590018 Karnataka, India

Major Project Phase II
Report On

"IOT-CONTROLLED AUTOMATED MULTI-FIELD IRRIGATION SYSTEM"

Submitted in partial fulfilment of the requirements of the award of the degree of

BACHELOR OF ENGINEERING
IN
ELECTRONICS AND COMMUNICATION ENGINEERING

Submitted by

1. ABHILASH N (4AI22EC002)

2. AMITH K R (4AI22EC006)

3. CHIRAAG A E (4AI22EC024)

4. MANASA T P (4AI22EC057)

Under the guidance of
Dr. Kumuda T
B.E., MTech., Ph.D., LMISTE., MIE.,
Professor
Department of ECE,
AIT, Chikkamagaluru.

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
ADHICHUNCHANAGIRI INSTITUTE OF TECHNOLOGY
CHIKKAMAGALURU, KARNATAKA, INDIA

2025-2026

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

ADHICHUNCHANAGIRI INSTITUTE OF TECHNOLOGY CHIKKAMAGALURU, KARNATAKA, INDIA

CERTIFICATE

This is to certify that Project work entitled *"IOT-Controlled Automated Multi-Field Irrigation System"* is a bonafide work carried out by **Abhilash N (4AI22EC002), Amith K R(4AI22EC006), Chiraag A E(4AI22EC024), Manasa T P(4AI22EC057)**, 7th Semester B.E. in partial fulfillment for the award of degree of Bachelor of Engineering in Electronics and Communication Engineering of the Visvesvaraya Technological University, Belagavi, during the year 2025-26. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report.

The project report has been approved as it satisfies the academic requirements of the prescribed for the said Degree.

1	GUIDE	Dr. Kumuda T B.E., MTech., Ph.D., LMISTE., MIE., Professor	
2	REVIEWER	Prof. Girish S B.E., MTech., (Ph.D.)., LMISTE Assistant Professor	
3	COORDINATOR	Dr. ANIL KUMAR C B.E., MTech., Ph.D., LMISTE., MIE Associate Professor	
4	HEAD OF THE DEPARTMENT	Dr. M A GOUTHAM B.E., MTech., Ph.D. Professor & Head	
5	PRINCIPAL	Dr. C.T Jayadeva B.E., MTech., Ph.D. Principal of AIT Chikkamagaluru	

External Viva

1	Examiner 1		
2	Examiner 2		

ACKNOWLEDGMENT

We express our humble Pranamas to his **holiness Parama Poojya Jagadguru Padmabhushana Sri Sri Sri Dr. Balagangadharanatha Mahaswamiji and Parama Poojya Jagadguru Sri Sri Sri Dr. Nirmalanandanatha Mahaswamiji** and also to **Sri Sri Gunanathana Swamiji** Sringeri Branch Chikkamagaluru who have showered their blessings on us for framing our career successfully.

We are deeply indebted to our honorable Director **Dr. C. K. Subbaraya** for creating the right kind of care and ambience.

We are thankful to our beloved Principal **Dr. C. T. Jayadeva** for inspiring us to achieve greater endeavors in all aspects of Learning.

We express our deepest gratitude to **Dr. M A Goutham**, Professor & Head, Department of Electronics & Communication Engineering for her valuable guidance, suggestions and constant encouragement without which success of our Project work would have been difficult.

We are grateful to our coordinators **Dr. Anil Kumar C,** for their excellent guidance, constant encouragement, support and constructive suggestions.

We are thankful to our guide **Dr. Kumuda T**, Professor, Dept. of Electronics & Communication Engineering, AIT, Chikkamagaluru for her inspiration and lively correspondence for carrying our project work.

We would like to thank our beloved parents for their support, encouragement and blessings.

And last but not least, I would be pleased to express our heartfelt thanks to all teaching and non-teaching staff of E&CE Department and our friends who have rendered their help, motivation and support.

ABHILASH N(4AI22EC002)

AMITH K R(4AI22EC006)

MANASA T P(4AI22EC057)

CHIRAAG A E(4AI22EC024)

ABSTRACT

This project presents the design and implementation of an IoT-enabled smart irrigation system that leverages GSM technology for remote control and monitoring of irrigation in multi-field agricultural setups. The system is centered on a microcontroller (STM32) interfaced with a SIM800L GSM module, which receives SMS commands from an Android application. These commands allow the farmer to remotely initiate irrigation for specific fields by specifying the field number and the desired irrigation duration. A relay module is used to control a water pump, while solenoid valves direct water to the appropriate field based on the command received. To prevent over-irrigation, soil moisture sensors monitor the moisture level during irrigation and can stop the process early if adequate moisture is detected.

Additionally, a dry run protection mechanism is integrated using a dry run sensor to detect water availability before activating the pump, thereby safeguarding the motor from damage due to dry running. The system also incorporates a timer-based control mechanism that automatically shuts off the irrigation process after the specified duration if the moisture condition is not met earlier. Upon completion or interruption, the system sends an SMS notification back to the user. This solution enhances water efficiency, ensures crop health, reduces manual labour, and provides a cost-effective and scalable smart farming tool suitable for resource-constrained environments.

DECLARATION

We declare that this project report titled "**IOT-Controlled Automated Multi-Field Irrigation System**" submitted in partial fulfillment of the degree of **B.E in Electronics and Communication Engineering** is a record of original work carried out by me under the supervision of **Dr Kumuda T**, and has not formed the basis for the award of any other degree, in this or any other Institution or University. In keeping with the ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

Signature of the student(s)

Name	**Signature**
1. Abhilash N	
2. Amith K R	
3. Chiraag A E	
4. Manasa T P	

TABLE OF CONTENTS

Chapter Title	Page No.
Acknowledgements	**i**
Abstract	**ii**
Declaration	**iii**
Contents	**iv**
List of Figures and List of Tables	**v**
CHAPTERS	**Page No.**
1. INTRODUCTION	01-10
1.1 Overview	01-02
1.2 Literature Survey	02-05
1.3 Comparison of State of Art	06-08
1.4 Problem Statement	09
1.5 Objectives	09
1.6 Proposed System	09-10
1.7 Scope of the Project	10
2. SYSTEM DESIGN	11-30
2.1 Block Diagram and Explanation	11-13
2.2 Hardware Requirements	13-27
2.3 Software Requirements	27-30
3. IMPLEMENTATION	31-43
3.1 Methodology	31
3.2 Circuit Diagram	32
3.3 Flow of Project	33-34
3.4 Algorithm for Automated Irrigation System	34-35
3.5 Test Cases	35-43
3.5.1 Hardware Implementation	35-39
3.5.2 Software Implementation	39-43
4. EXPERIMENT RESULTS AND DISCUSSION	44-45
4.1 Advantages	44
4.2 Applications	45
CONCLUSION AND FUTURE SCOPE	46
Conclusion	46
Future Scope	46
REFERENCE	47
APPENDIX	48-49

LIST OF FIGURES

Figure No	Figure Name	Page No.
Figure 2.1	Block Diagram of an Automated Irrigation System	11
Figure 2.2	STM32F103C8T6	15
Figure 2.3	SIM900A GSM Module	16
Figure 2.4	4-Channel Relay Module	17
Figure 2.5	6V DC Water Pump	18
Figure 2.6	Solenoid Valve	20
Figure 2.7	16x2 LCD Display	21
Figure 2.8	Soil Moisture	24
Figure 2.9	Dry Run Sensor	25
Figure 2.10	ESP-32 Microcontroller	26
Figure 3.1	Circuit Diagram for Automated Irrigation System	32
Figure 3.2	Flowchart for Automated Irrigation System	33
Figure 3.3	System Initialization and Processing	36
Figure 3.4	Display Status of Motor and Valves on 16x2 LCD Display	37
Figure 3.5	Snapshot of SMS Command	39
Figure 3.6	Snapshot of MIT app	42

LIST OF TABLES

Figure No	Figure Name	Page No.
Figure 1.3	Comparison of State of Art	06-08

CHAPTER 1

INTRODUCTION

1.1 Overview

Agriculture continues to remain the backbone of many economies, particularly in developing nations like India, where a significant portion of the population depends on farming for livelihood. Despite its importance, the sector still relies heavily on traditional irrigation practices, which are often inefficient, labor-intensive, and prone to significant water wastage. With increasing water scarcity and rising demand for food production, there is an urgent need to adopt automated irrigation solutions that not only save water but also improve crop yield and reduce the dependence on manual supervision.

Water management plays a pivotal role in modern agriculture. In many regions, especially those prone to erratic rainfall patterns or seasonal droughts, efficient water distribution is a challenge. Disputes often arise among farmers over the allocation of irrigation water, further complicating agricultural management. In traditional setups, farmers are required to visit their fields physically to open or close irrigation gates, monitor water flow, and manually regulate the pump. This process is not only time-consuming but also demands substantial physical effort. At times, farmers must travel long distances repeatedly just to check soil conditions or turn the pump on and off, resulting in wasted time and energy.

The introduction of automation in irrigation systems offers a promising solution to these challenges. By using IoT-enabled smart systems, irrigation can be automated to open and close water gates according to real-time crop requirements and soil moisture conditions. These systems can also be controlled remotely through SMS commands, mobile applications, or even web-based platforms, giving farmers the ability to manage multiple agricultural fields from a distance. Such a capability is especially useful for those cultivating crops across scattered plots of land, as it eliminates the need for continuous on-site supervision. One of the notable features of modern smart irrigation systems is dry run protection. This mechanism detects whether there is adequate water supply before activating the pump. If no water is detected, the system prevents the motor from running in a dry state, thereby safeguarding it from potential overheating, damage, or energy wastage. Another essential feature is the programmable timer-based irrigation cycle, which allows farmers to pre-set specific watering durations for each field. This ensures that crops receive adequate hydration without the risk of over-irrigation, helping conserve water while maintaining soil health.

However, the adoption of such technologies faces certain barriers in rural areas. Many farmers, particularly in remote villages, lack awareness or technical knowledge regarding the use of IoT-based systems. Limited education, insufficient training, and economic constraints often restrict the access of small-scale farmers to these advanced tools. As a result, they continue to rely on conventional irrigation

methods despite the inefficiencies. Overcoming these challenges requires affordable, user-friendly, and easily scalable solutions that can work even in areas with poor internet connectivity.

This project introduces an IoT-enabled smart irrigation system, specifically designed to address these issues. Unlike many solutions that require internet connectivity, this system is controlled via SMS-based communication, making it highly suitable for rural and resource-constrained regions. It enables farmers to manage irrigation for multiple fields individually, depending on crop type, soil condition, and moisture requirements. This field-specific management ensures optimal water distribution, minimizes wastage, and promotes better crop productivity.

By combining dry run protection, soil moisture monitoring, and timer-based irrigation scheduling, the system offers a reliable and sustainable approach to water management in agriculture. Farmers are relieved from the burden of constant field visits and manual supervision, as the system automatically ensures that water is supplied only when and where it is required. Ultimately, this solution not only enhances water-use efficiency but also contributes to improved crop yield, reduced labor costs, and long-term sustainability in agriculture.

1.2 Literature Survey

(Abdelmadjid Saadet al. 2020) Having the ability to monitor water via sensors provides farmers the power to enhance the growth of the crops. The use of these sensors provides accurate and real-time information on different parameters related to water so that farmers can effectively intervene at the right moment. Having the ability to monitor water via sensors provides farmers the power to enhance the growth of the crops. The use of these sensors provides accurate and real-time information on different parameters related to water so that farmers can effectively intervene at the right moment.By this paper we conclude that enabling real-time monitoring through sensors, farmers can make timely and informed decisions, ensuring optimal water usage. This leads to healthier crops, increased yield, and more sustainable agricultural practices. [8]. (Srinath M S 2014) Agriculture is the cornerstone of many developing economies, such as India, where a large percentage of the population relies on farming. However, traditional irrigation methods often lead to excessive water consumption, inefficient resource usage, and high labor dependency. In rural areas, the lack of education, awareness, and access to modern technologies further hampers the adoption of efficient water management practices. Problems such as continuously running motors, manual supervision of irrigation gates, and poor scheduling result in significant water wastage and disputes over water distribution. Based on the above technological paradigms, existing studies are selected for a general review regarding the application of these technologies for water management in the agriculture field. Existing research highlights the potential of IoT and automation technologies in addressing water management challenges in

agriculture. These solutions offer efficient, data-driven irrigation methods that conserve water and support sustainable farming practices. [7]. (Sreekanth Narendran et al.2015) The server collects the water monitoring data forwarded by the gateway and store in a database for analytics as well as displaying them in web-based dashboard. Data collection for the system is configured in terms of timestamp, days and week in the web-based dashboard. The advent of the Internet of Things (IoT) has revolutionized agriculture by introducing intelligent, automated solutions that optimize water usage and improve crop productivity. The server stores water monitoring data from the gateway and displays it on a web-based dashboard with time stamped tracking. IoT integration enables automated, data-driven irrigation, improving efficiency and crop productivity. As referred in the paper, the server collects water monitoring data from the gateway, stores it in a database, and displays it through a web-based dashboard with time stamped logs. This IoT-based approach facilitates automated, data-driven irrigation, significantly improving water efficiency and crop productivity. [1], (Thinagaran Perumal et al.2017) Integrated in their system to provide a sustainable environment. The paper titled Internet of Things for Smart cities provide an insight into realization of IoT network along with the required back-end network and services. The paper also describes the protocol stack used in the architecture. Micro Water Distribution Networks describes about sustainably building a water distribution system in Indian villages, considering the local factors like labour, local community needs, climate and time for implementation. The paper gives an idea about the various challenges that had to be faced while building such a system in India. An Integrated System for Regional Environmental Monitoring and Management Based on Internet of Things is also a similar work incorporating IoT, cloud computing, global positioning system etc. to monitor the environment. They have put forward an architecture that can handle huge data. The reviewed papers highlight the effective integration of IoT, cloud computing, and localized design in building sustainable water and environmental monitoring systems. These insights support the development of region-specific, data-driven solutions for smart agriculture and rural infrastructure [2]. (Gilroy P.Pereira et al. 2023) IoT-based smart irrigation systems integrate sensors, wireless networks, and microcontrollers to monitor environmental parameters—such as soil moisture, temperature, humidity, and water flow—in real time. These systems utilize components like ESP32 microcontrollers, ultrasonic sensors, flow meters, solenoid valves, and cloud-based dashboards, enabling farmers to manage irrigation remotely via mobile apps or SMS, even in regions with limited internet connectivity. We can use the Internet of Things (IoT) in any application that requires data collection, automation or control. With the growing popularity of IoT, there has been an increase in the ideas surrounding smart agriculture technology. This smart IoT-enable drip irrigation system using an ESP32 microcontroller. The system comprises an ESP32, solenoid valve, soil moisture sensor, temperature sensor, air humidity sensor, and water flow sensor. The referenced paper demonstrates an IoT-enabled drip irrigation system using ESP32 and various sensors to automate and optimize water

usage. It concludes that such systems are effective in enhancing agricultural productivity, especially in resource-constrained environments [6]. (SHEREEN ISMAIL et al 2022) Systems are capable of automatically triggering irrigation when soil moisture falls below a predefined threshold and halting it once sufficient water has been dispensed, thereby preventing over-irrigation and conserving energy. The system automatically initiates irrigation when soil moisture drops below a set level and stops it once optimal moisture is reached. This prevents over-irrigation, conserves water, and reduces energy consumption efficiently. As referenced in the paper, the system triggers irrigation based on soil moisture thresholds and stops when optimal levels are reached. This ensures efficient water use, prevents over-irrigation, and supports sustainable farming practices. Some advanced implementations also incorporate features such as automated dry run protection for pumps, water level monitoring in storage tanks, and turbidity detection to initiate purification processes. In the end, it proposes the requirement for the openness of data resources for academic research and business ventures in the agricultural domain. Advanced systems include features like dry run protection, water level monitoring, and turbidity-based purification to enhance reliability and efficiency. The paper concludes by emphasizing the need for open access to agricultural data to support research and innovation[4] .(Kurniawan D. Irianto et al. 2022) Humans still operate irrigation systems in agriculture manually, such as opening and closing floodgates. However, in the current era of technological advances, it is possible to make innovations by making irrigation systems smarter with the help of IoT. In some rural setups, systems have even been designed to cut off water flow across entire zones, like villages, to prevent large-scale wastage. Subsurface irrigation, the irrigation system irrigates the plants through pipes that have been planted in the roots so that water can go directly to the plant roots. Drip irrigation, the system uses a pipe with small holes so that the water that comes out in the form of droplets will go directly to the plants. This paper clearly gives the water management in different methods. The paper highlights the transition from manual to smart irrigation systems using IoT, enabling efficient water control even in rural areas. It also explores methods like subsurface and drip irrigation, emphasizing targeted water delivery and improved water management practices. [5]. The implementation of these advanced systems is not without challenges. Illiteracy, low digital literacy, infrastructural constraints, and the rural-urban technology divide significantly hinder widespread adoption [8]. They also proposed a remote automated irrigation system, which provides notifications via mobile SMS using fog and IoT cloud technologies. A mathematical model to predict water amounts needed for irrigation was proposed in addition to, a real-time monitoring of agricultural parameters including water. For more reliable and intelligible operation, the authors in demonstrated a simple and low-cost IoT control system with multiple control capabilities. The proposed system in comprises three different controllers . To overcome these barriers, researchers and practitioners emphasize the need for user-friendly interfaces, localized training, and supportive policies or subsidies from government and agricultural institutions.

The paper proposes a remote IoT-based irrigation system using fog and cloud technologies, offering SMS notifications and predictive water modeling. It also highlights the importance of user-friendly, low-cost designs and stresses the role of training and policy support to ensure successful adoption in agriculture. [4].(RITESH KUMAR SINGH et al. 2021).It explores the major components of IoT that can be instrumental for smart farming, along with the introduction to the various network topologies and network architectures mapped with the layers of IoT. Ultimately, integrating IoT, wireless sensor networks (WSNs), and cloud computing into agricultural water management creates a robust framework for sustainable farming. It explores the major components of IoT that can be instrumental for smart farming, along with the introduction to the various network topologies and network architectures mapped with the layers of IoT.The paper explores key IoT components essential for smart farming, including network topologies and architectures aligned with IoT layers. By integrating IoT, WSNs, and cloud computing, it establishes a strong foundation for efficient and sustainable agricultural water management.[3]. These technologies enable precision agriculture by automating irrigation, minimizing manual labor, and supporting real-time decision-making, which is crucial in addressing the growing demand for food and water security worldwide [1].

1.3 Comparison of State of Art

SL.NO	PAPER TITLE	NAME OF THE AUTHORS	TECHNICAL IDEAS / ALGORITHMS USED IN THE PAPER	SHORTFALL/ DISADVANTAGES
1.	An Internet of Things (IoT) based Sustainable Water Management	Sreekanth Narendran, Preeja Pradeep, Maneesha Vinodini Ramesh	IoT-based architecture using layered design (Things, Network, Middleware Application), Sensors (flow meters, pressure sensors, ultrasonic rangefinders), GSM for communication, REST ful APIs, Node.js for data processing	Power outages in rural areas, Lack of awareness and education, High manual effort in water collection, Water wastage due to continuous motor operation, Limited internet connectivity in rural areas.
2.	Internet of Things (IoT) Enabled Water Monitoring System	Thinagaran Perumal1 , 1 Md Nasir Sulaiman, 2 Leong.C.Y	Arduino UNO, ESP826 Wi-Fi module, sensors for pH, temperature; cloud integration via ThingSpeak; real-time data monitoring through mobile/ app.	Limited scalability, No predictive analytics, Dependent on stable internet, Basic data security.

3.	AgriFusion: An Architecture for IoT and Emerging Technologies Based on a Precision Agriculture Survey	Ritesh Kumar Singh, Rafael Berkvens, Maarten Weyn	IoT, Wireless Sensor Networks, Machine Learning, Artificial Intelligence, Software Defined Networks, UAVs, Remote Sensing, Energy Harvesting.	High data volume and latency, Limited energy resources in sensor nodes, Security and privacy risks in smart farming.
4.	IoT-Based Water Management Systems: Survey and Future Research Direction	Shereen Ismail, Diana W. Dawoud, Nadhem Ismail, Ronald Marsh, Ali S. Alshami	IoT, Wireless Sensor Networks (WSN), Optical Sensing, Multispectral Imaging, Machine Learning (ML), Cloud Computing, Edge Computing, Smart Meters.	Limited research in oilfield applications Harsh environments (high pressure/ temp/ corrosion) High cost and size of multispectral sensors Communication challenges in underground settings
5.	Design of Smart Farm Irrigation Monitoring System Using IoT and LoRa	Kurniawan D. Irianto Jurnal	IoT sensors (moisture, temperature, humidity), Arduino/ESP8266, Cloud storage, Decision Tree ML model for soil moisture prediction, Automation, Mobile app for monitoring	Limited scalability for large farms, Requires reliable internet, Power supply challenges, Decision tree may lack accuracy in complex conditions

6.	IoT-Enabled Smart Drip Irrigation System Using ESP32	Gilroy P.Pereira , Mohamed Zachary, Fawwad Daroge MDPI	IoT, Smart sensors, Cloud-based monitoring, Mobile application interface, Real-time data acquisition, Data analysis using microcontrollers and sensor modules	Lack of continuous water quality monitoring, Delayed response to contamination, Manual sampling prone to errors, Inadequate infrastructure in rural areas
7.	GSM based Automatic Irrigation Control System for Efficient Use of Resource and Crop Planning by Using an Android Mobile.	Pavithra D. S, M S. Srinath	IoT-based system using Arduino UNO, GSM Module, Soil Moisture Sensor, Water Pump Control	Manual irrigation systems lead to water wastage, no automation, inefficient use of water resources, no real-time monitoring
8.	Water Management in Agriculture: A Survey on Current Challenges and Technological Solutions	Abdelmadjid Saad, Abou El Hassan Benyamina, Abdoulaye Gamatie	IoT, Wireless Sensor Networks (WSN), Cloud Computing, Cyber Ph- -ysical Systems (CPS), Remote Sensing, Machine Learning, GIS, MQTT/CoAP protocols	High cost and delay in real-time monitoring, Difficulty in detecting pipeline leaks, Water wastage in traditional irrigation, Health risks from using recycled water, High energy consumption of sensing systems.

1.4 Problem statement

In multi-field farming, effective irrigation management is critical to conserving water, reducing labour, and improving crop productivity. However, traditional irrigation systems lack automation, real-time monitoring, and the flexibility to manage multiple zones independently. Farmers often struggle with issues like dry running of pumps, which can damage equipment and lead to increased maintenance costs. Manual irrigation not only consumes significant time and labour but also results in inefficient water usage due to the inability to schedule watering based on field-specific requirements. These challenges are further amplified in remote or large-scale agricultural settings where constant physical supervision is not feasible.

There is a pressing need for a smart irrigation solution that integrates real-time field monitoring, automated timer-based control, and protection mechanisms such as dry run detection. The absence of such a system leads to water wastage, energy loss, and reduced crop health. An IoT-enabled, SMS-controlled smart irrigation system can address these limitations by allowing farmers to remotely control and monitor irrigation schedules, automate watering times for different fields, and prevent pump failures due to dry running. Such a system ensures optimal resource use, minimizes manual effort, and enhances productivity in multi-field farming environments

1.5 Objectives

- To use timer-based scheduling to optimize irrigation duration for different fields.
- To enable individual field management for multi-zone farms.
- To minimize manual labor and reduce the time farmers spend supervising irrigation.
- To reduce water wastage through precise and need-based irrigation.
- To display system information and sensor data on a local 16x2 LCD display.
- To utilize the STM32F103C8T6 microcontroller for efficient and low-power system control.
- To support real-time feedback and alerts to farmers via SMS for system status and faults.
- Design a user-friendly Android application for farmers using MIT App Inventor.

1.6 Proposed system

An IoT-enabled smart irrigation system for multi-field farming integrates soil moisture sensors, weather data, and SMS-based control to automate and optimize water usage across different fields with varying crop needs. The system will continuously monitor soil moisture levels in each field to determine precise irrigation schedules, activating water pumps and valves as required. Farmers can remotely manage and override automated schedules, as well as receive system status updates and alerts,

via SMS commands. The system incorporates automated dry run control to prevent pump damage by detecting and halting operation in the absence of water flow. Timer management will allow for setting specific irrigation durations for each field, ensuring efficient water distribution based on individual crop requirements and growth stages, all controllable through simple SMS commands.

A central microcontroller (such as STM32) interfaces with sensors and a GSM module to manage communication and system operations. Timer-based control enables automated irrigation cycles according to the specific water requirements of each field. To protect the irrigation pump, a dry run detection mechanism is integrated using water flow or current sensors, which shuts off the pump automatically if water supply is insufficient, thus preventing damage. This system reduces manual labor, prevents water and energy wastage, and ensures that each field receives the right amount of water at the right time, leading to healthier crops and improved farming efficiency. Overall, this system significantly reduces manual labor, improves water use efficiency, and enhances crop health. With its modular, scalable, and cost-effective design, the proposed solution is well-suited for both small-scale and large-scale agricultural applications, promoting sustainable and intelligent farming practices.

1.7 Scope of the Project

The proposed IoT-Controlled Automated Multi-Field Irrigation System has a wide scope in modern agriculture, particularly in regions facing water scarcity and labour shortages. By integrating soil moisture sensing, GSM communication, and timer-based automation, the system ensures that irrigation is carried out only when necessary, reducing water wastage and enhancing crop yield. The scope extends to both small-scale farms and large agricultural setups, as the modular design allows additional fields and sensors to be integrated easily.

The system is especially suitable for rural areas with poor internet connectivity, since it operates via SMS commands, making it more accessible to farmers with basic mobile phones. Features like multi-field management**,** dry run protection**,** and real-time feedback through SMS alerts add reliability and safety to irrigation practices. In addition, the use of a mobile app built with MIT App Inventor provides a user-friendly interface, minimizing the need for technical knowledge.

Future extensions of the project may include integration with IoT cloud platforms for data logging, predictive irrigation based on weather forecasts, and the use of renewable energy sources like solar power to drive the pump. Thus, the project not only addresses immediate irrigation challenges but also contributes toward sustainable agriculture and the efficient use of natural resources.

CHAPTER 2

SYSTEM DESIGN

The system design combines an STM32 microcontroller and an ESP32 module to manage multi-field irrigation efficiently. The STM32 controls sensors, pumps, and valves, while the ESP32 enables IoT connectivity and MIT App communication. Together, they provide SMS and app-based monitoring, automated timing, and dry-run protection for smart irrigation control.

2.1 Block Diagram & Explanation

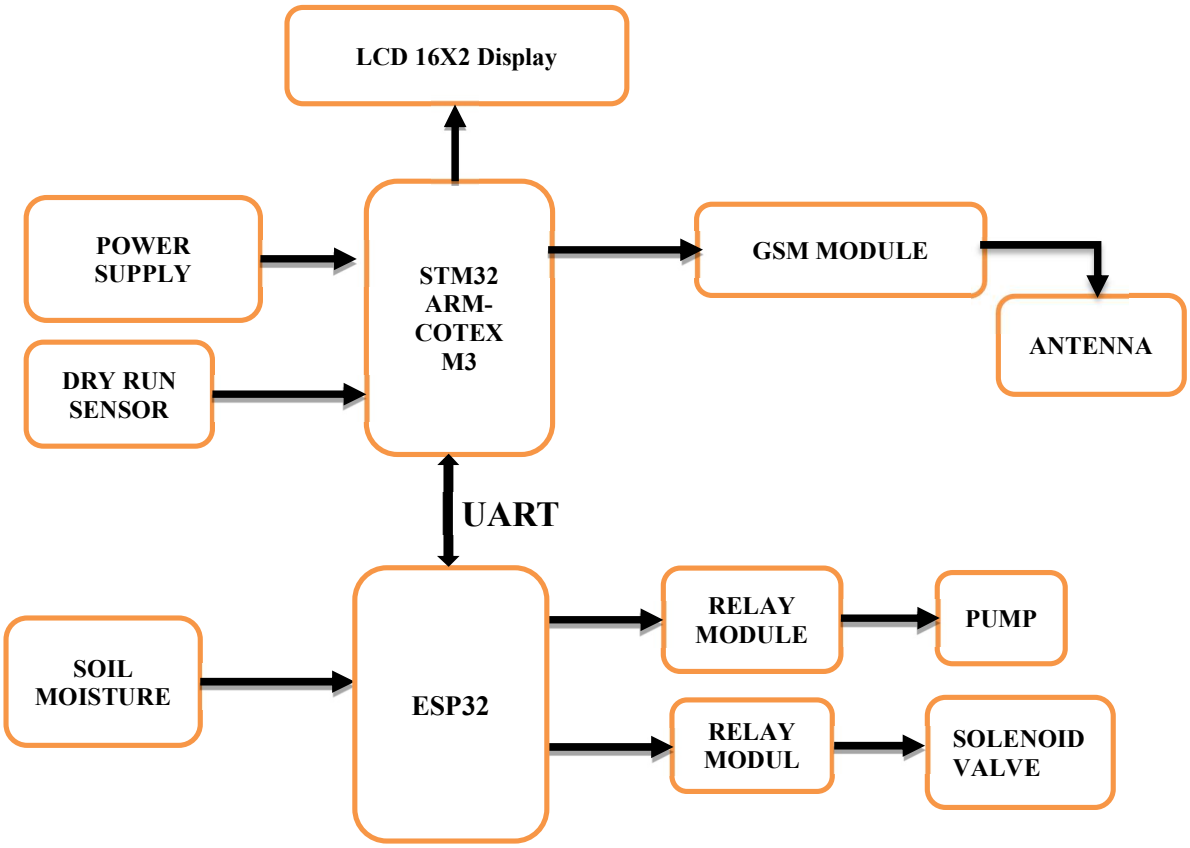

Fig 2.1 Block Diagram of an Automated Irrigation System

1. **STM32 ARM Cortex-M3 (Microcontroller)**

- The STM32 is the **brain of the system**.
- It collects data from sensors (soil moisture, dry run), processes SMS commands from the GSM module, and controls the relay/motor driver for irrigation.
- It executes decision-making: when to start/stop the pump, which solenoid valve to activate, and for how long irrigation should run.

2. **ESP32 (Microcontroller)**

- ESP32 reads soil-moisture sensors, processes the data, and decides when irrigation is needed
- It sends and receives commands through SMS/IoT using GSM or Wi-Fi to update irrigation status.
- ESP32 operates the relay, pump, and solenoid valves based on sensor readings or user commands

3. **GSM Module (SIM800L) + Antenna**

- Provides wireless communication through SMS.
- Farmers send commands (e.g., Field 2 ON for 10 minutes) via SMS.
- The module transmits this command to the STM32 and sends back acknowledgments or alerts to the user after irrigation is complete.
- The antenna ensures strong network reception for reliable communication.

4. **Motor Driver + Pump**

- Since the STM32 cannot directly drive high-power devices, a motor driver/relay module is used as an interface.
- The pump supplies water from the source (tank/borewell) when switched on.
- It operates only if dry run sensor conditions are satisfied (to protect the motor).

5. **Solenoid Valves**

- These valves control which field receives water**.**
- Based on the farmer's SMS command or sensor data, the STM32 opens the corresponding solenoid valve, ensuring precise water delivery to the selected zone.

6. **Soil Moisture Sensor**

- Continuously measures the water content in the soil.
- If the soil already has sufficient moisture, the controller can skip or stop irrigation, thus preventing overwatering.
- Data is also useful for timer adjustment and efficient water scheduling.

7. **Dry Run Sensor**

- Detects whether water is present in the source before activating the pump.
- If no water is detected (dry condition), the STM32 prevents the pump from turning on, avoiding motor damage.

8. **DHT11 (Temperature and Humidity Sensor)**

- Monitors environmental conditions like air temperature and humidity.
- Provides supporting data to optimize irrigation cycles according to weather conditions.
- For example, in high humidity, irrigation can be reduced to save water.

9. **Power Supply Unit**

- Provides regulated DC power to all components (STM32, GSM module, sensors, relays, valves).
- Typically derived from an AC supply or a battery/solar source for rural setups.

The block diagram of the IoT-Controlled Automated Multi-Field Irrigation System is designed to make the process of watering multiple agricultural fields easier, smarter, and more efficient. At the centre of this system is the STM32 ARM Cortex-M3 microcontroller, which acts as the main controller that receives information from different sensors such as the soil moisture sensor, dry run sensor. Using this data, along with commands received through SMS from the GSM module, the controller decides when and where irrigation should take place. The GSM module plays an important role in allowing two-way communication between the farmer and the system. Farmers can send SMS commands to control irrigation and also receive real-time alerts about soil moisture, pump status, and other conditions. The motor driver circuit controls the water pump, while solenoid valves direct water to the particular field that needs it. This ensures that each field gets the right amount of water at the right time, reducing wastage and improving efficiency.

The system also includes several safety and efficiency features. The dry run sensor protects the pump by stopping it from running when there is no water in the source tank. The soil moisture sensor helps in saving water by ensuring irrigation happens only when the soil is dry. A regulated power supply provides stable power to all components, and a backup battery ensures the system works even during power cuts. Solar power can also be integrated for greater sustainability. Additionally, the system can include an LCD display or IoT-based dashboard for local monitoring, and an EEPROM can be used to store sensor data. Overall, this project offers a practical, automated, and SMS-controlled irrigation solution that saves time, conserves water, reduces manual labour, and supports sustainable multi-field farming practices. the modular design of the system allows it to be easily expanded to manage additional fields or different types of crops. Each field can be independently controlled based on its moisture requirements, optimizing water usage and preventing over-irrigation. The combination of sensors, microcontroller logic, and automatic control not only saves water but also reduces energy consumption by running the pump only when needed. By implementing features such as dry run protection and data logging, the system enhances the safety and performance of irrigation operations. Overall, this project demonstrates how IoT and embedded systems can be effectively used to modernize agriculture, making farming more sustainable, efficient, and productive.

2.2 Hardware Requirements

The proposed irrigation system is built using a combination of sensing, control, and communication hardware, all integrated to automate field irrigation effectively. At the heart of the setup is the STM32F103C8T6 microcontroller, commonly known as the "Blue Pill," which serves as the main controller. It is responsible for receiving commands, processing sensor data, and controlling the pump, solenoid valves, and display. To enable remote communication, a SIM900A GSM module is used. This module allows the farmer to send and receive SMS commands, making the system suitable even in rural areas where internet connectivity is limited. An external antenna is connected to the GSM module to ensure strong signal reception for reliable operation. For switching high-power devices like the pump and solenoid valves, a relay module is included. The relay acts as an electronic switch that isolates the low-power control circuit from the high-power irrigation equipment, ensuring safe and efficient operation.

The water pump is used to draw water from the source and deliver it to the fields. It is controlled by the relay and is capable of providing enough flow and pressure for irrigation. To distribute this water to specific sections of the field, solenoid valves are used. Each solenoid valve directs the flow of water to a particular zone, allowing selective irrigation depending on the crop requirements. For real-time system monitoring, a 16x2 LCD display is included. This display shows important messages such as pump status, valve status, or fault alerts, enabling easy local monitoring without requiring external devices.

The entire setup is powered by a dedicated power supply unit**,** which converts the input power into stable voltages suitable for different components like the microcontroller, GSM module, and sensors. To ensure compatibility between components that operate on different voltage levels, a logic level shifter may be used when necessary. Supporting components such as wires, connectors, and a PCB or breadboard are used for interconnections, prototyping, and mounting. Together, these hardware units create a reliable, modular, and farmer-friendly irrigation system capable of automating multi-field water management with minimal human intervention.

In this project, the hardware components used are:

1. STM32F103C8T6 Microcontroller (Blue Pill)
2. SIM900A GSM Module
3. Relay Module (4-Channel)
4. Water Pump (12V DC)
5. Solenoid Valves (3 Nos)

6. 16x2 LCD Display
7. Power Supply Unit
8. Wires, Connectors, PCB/Breadboard
9. Soil Moisture
10. Dry Run Sensor
11. ESP32

2.2.1 STM32F103C8T6 ("Blue Pill")

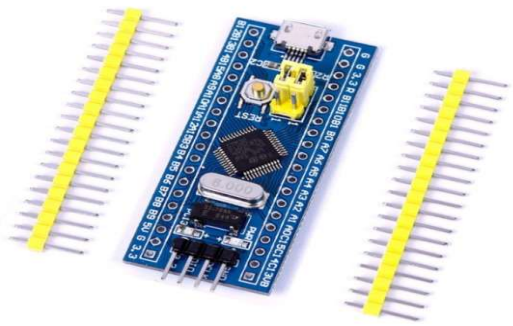

Fig 2.2 STM32F103C8T6

Overview:

The STM32F103C8T6, popularly known as the "Blue Pill," is a powerful, low-cost microcontroller development board. It is based on the ARM Cortex-M3 32-bit processor and offers excellent performance compared to traditional microcontrollers like Arduino Uno. It is commonly used in embedded system applications, IoT projects, robotics, and industrial automation.

Working Principle:

The microcontroller functions as the “brain” of the project. It receives input data from sensors (such as the soil moisture sensor), processes this information using programmed logic, and generates control signals for actuators (solenoid valve or water pump). It operates on digital logic and can interface with multiple analog and digital devices.

Technical Specifications:

- Processor: ARM Cortex-M3, 32-bit, up to 72 MHz clock speed
- Flash Memory: 64 KB
- SRAM: 20 KB
- Operating Voltage: 3.3V (with 5V tolerant pins)
- Clock: 8 MHz external crystal + 72 MHz PLL

- Peripherals:
- 12-bit ADC (10 channels)
- PWM outputs
- Communication interfaces: USART, SPI, I2C, USB

Advantages:

- Higher processing speed compared to Arduino boards
- Cost-effective with rich features
- Compact and energy-efficient
- Multiple communication and interfacing options

Role in Project:

The STM32 board acts as the central controller of the irrigation system. It collects soil moisture readings, compares them with the threshold value, and controls water supply by activating the solenoid valve or pump.

2. 2.2 SIM900A GSM Module:

Fig 2.3 SIM900A GSM Module

Overview:

The SIM900A is a compact GSM/GPRS module widely used in IoT and embedded applications for wireless communication. It supports SMS, voice calls, and GPRS data transmission, making it ideal for projects that require remote monitoring and control. Due to its small size and low power consumption, it is well-suited for mobile and rural applications where internet connectivity is limited.

Working principle:

The SIM900A module works by connecting to the GSM cellular network using a SIM card. It communicates with the STM32 microcontroller via the UART interface, allowing the system to send and receive SMS commands. In this project, farmers can control irrigation by sending SMS messages, while the system responds with status updates such as pump ON/OFF, water availability, or completion of irrigation. An external antenna is used to enhance network reception for reliable communication.

Technical Specifications:

- Quad-band GSM/GPRS (850/900/1800/1900 MHz)
- Supports SMS, voice calls, and GPRS data

- Communication interface: UART (AT Commands)
- Compact design with low power consumption
- Requires external antenna for stable signal

Advantages:

- Reliable SMS-based communication even in areas without internet
- Low cost and compact, making it easy to integrate
- Supports wide GSM frequency bands for global use
- Works with simple AT commands, ensuring easy programming

Role in Project:

The SIM900A module enables wireless communication between the farmer and the irrigation system. Farmers can send SMS commands to start irrigation for a particular field or set a timer. The system also sends alerts, such as “Pump ON,” “Field irrigated,” or “Dry run detected,” ensuring the farmer is informed in real-time. This feature makes the system practical for rural regions with poor internet connectivity but strong GSM coverage.

2.2.3. 4-Channel Relay Module:

Fig 2.4 4-Channel Relay Module

Overview:

A relay module is an electromechanical switch that allows a low-power microcontroller signal to control high-power electrical devices. The 2-channel relay module used in this project provides two independent switching channels, enabling control of devices such as the water pump and solenoid valves. It is commonly used in home automation, industrial automation, and IoT projects where safe isolation between control and power circuits is required.

Working Principle:

The relay module operates by receiving a control signal from the STM32 microcontroller. When activated, the relay coil energizes, creating a magnetic field that pulls the switch contacts, allowing current to flow to the connected high-power device. When the signal is turned off, the contacts return to their default position, cutting off power. This switching mechanism ensures that the microcontroller, which operates at low voltage, can safely control high-current loads such as pumps and valves.

Technical Specifications:

- Two independent relay channels
- Opto-isolated inputs for safe operation
- Capable of switching AC and DC loads
- Control input compatible with microcontroller logic levels
- LED indicators for relay status

Advantages:

- Provides electrical isolation between control and load circuits
- Can safely switch high-power devices using low-power signals
- Reliable and durable for continuous operation
- Simple interface with microcontrollers

Role in Project:

In this irrigation system, the 2-channel relay module is used to control the water pump and solenoid valves. The STM32 microcontroller sends control signals to the relay, which then switches the pump ON/OFF and directs water to the appropriate field through the solenoid valves. This allows automated and safe control of high-power irrigation equipment directly from the microcontroller.

2.2.4 12V DC Water Pump

Fig 2.5 12V DC Water Pump

Overview

A DC water pump is an electromechanical device that converts electrical energy into hydraulic energy to move water from a source to the desired location. The 12V DC pump used in this project is compact, reliable, and commonly employed in small-scale irrigation, aquaponics, and household water circulation systems. It is well-suited for projects requiring controlled water delivery.

Working Principle

The pump operates when a DC voltage is applied to its motor, causing the impeller inside the pump to rotate. This rotation creates a pressure difference, drawing water from the inlet and pushing it out through the outlet. In this irrigation system, the STM32 microcontroller controls the pump indirectly through the relay module. When irrigation is required, the relay activates the pump, supplying water to the solenoid valves, which then direct the flow to the selected field. To maintain stable performance, the system can monitor pump runtime, temperature, or current flow, helping to detect issues like dry running, pipe blockages, or motor overheating. If any fault is detected, the controller can immediately stop the pump to avoid damage. This combination of automated control and protective measures ensures reliable water delivery, reduces maintenance needs, and enhances the overall efficiency of the irrigation setup.

Technical Specifications

- Operates on 12V DC supply
- Current requirement suitable for small to medium irrigation tasks
- Provides steady water flow for controlled irrigation
- Compatible with relay-based switching
- Offers good durability with a robust motor suitable for outdoor conditions
- Delivers consistent pressure output for efficient water distribution

Advantages

- Simple and reliable operation
- Compact and portable, suitable for small irrigation setups
- Energy-efficient and easy to control using a microcontroller and relay
- Provides sufficient pressure for multi-field irrigation through solenoid valves

Role in Project:

The 12V DC water pump is the core actuator responsible for delivering water to the fields. Controlled by the relay and STM32 microcontroller, it pumps water from the storage tank or source into the irrigation system. The pump operates only when required, based on soil moisture levels or

timer settings, ensuring efficient water usage and preventing wastage. It is also protected through dry-run detection, which prevents the motor from running without water. The system ensures stable pressure and consistent flow across multiple fields. Its compact size, reliability, and compatibility with battery or solar power make it ideal for remote agricultural applications.

2.2.5 Solenoid Valves (3 Nos)

Fig 2.6 Solenoid Valves

Overview

A solenoid valve is an electromechanical device used to control the flow of liquids or gases. In irrigation systems, it functions as an automatic gate that opens or closes the water flow based on control signals from the microcontroller.

Working Principle

When electrical power is applied, the solenoid coil inside the valve generates a magnetic field. This magnetic force moves a plunger, which opens the valve and allows water to flow. When power is removed, the plunger returns to its original position, closing the valve and stopping water flow.

Technical Specifications

- Type: Normally Closed (NC)
- Operating Voltage: 24V DC
- Pressure Range: 0 – 0.7 MPa
- Orifice Size: 2.5 mm
- Control Method: On/Off by electrical signal

Advantages

- Provides reliable and quick control of water flow
- Can handle relatively high-water pressure

- Simple ON/OFF operation with relay or MOSFET control
- Durable and long-lasting

Role in Project

In this irrigation system, the solenoid valves regulate water supply to different fields. The STM32 microcontroller activates a particular valve when irrigation is required and deactivates it once the soil moisture level reaches the desired threshold or the timer ends. This allows efficient distribution of water to selected zones.

2.2.6. 16x2 LCD Display (Standard or I2C)

Fig 2.7 16x2 LCD Display (Standard or I2C)

Overview

A 16x2 LCD (Liquid Crystal Display) is a simple alphanumeric display module that can show 16 characters per row across 2 lines. It is widely used in embedded systems and IoT applications to provide real-time information in a readable format.

WorkingPrinciple

The LCD works by controlling liquid crystals using electrical signals to block or allow light, forming visible characters on the screen. It can operate in either 4-bit or 8-bit mode, where the microcontroller sends data and control signals to display text messages. In this system, the STM32 microcontroller updates the LCD with status information.

Technical Specifications

- Display format: 16 characters × 2 lines
- Interface: 4-bit/8-bit parallel or I2C
- Character type: Alphanumeric (ASCII)
- Backlight for visibility in low light conditions

Advantages

- Low cost and easy to interface with microcontrollers
- Provides clear, real-time system feedback
- Can be operated in parallel or I2C mode for flexibility
- Compact and lightweight for integration in embedded projects

Role in Project:

In this irrigation system, the 16x2 LCD display is used to show important system messages such as *"Pump ON"*, *"Valve 1 Active"*, or *"Irrigation Complete"*. This feature allows farmers or technicians to locally monitor the system without needing a mobile phone or additional devices.

2.2.7 Power Supply Unit

Overview

The power supply unit provides the required electrical energy for all components of the irrigation system. Since different modules operate at different voltage levels, the power supply ensures that stable and regulated voltages are delivered to the microcontroller, GSM module, relays, and sensors.

Working Principle

The system uses a main DC adapter as the input power source. This supply is stepped down and regulated using voltage regulators or buck converters to provide the correct operating voltages. For example, the STM32 microcontroller requires 3.3V, the GSM module operates around 4V, and the relays and solenoid valves typically need 12V. The power supply unit ensures safe and efficient distribution of power to each part of the circuit.

Technical Specifications

- Input: 12V DC adapter (sufficient current rating)
- Output: Regulated 12V, 5V, and 3.3V DC
- Components: Voltage regulators or buck converters (e.g., LM2596, AMS1117)

Advantages

- Provides stable and regulated voltage levels
- Protects components from overvoltage or power fluctuations
- Ensures reliable and continuous operation of the system
- Can supply multiple components simultaneously

Role in Project;

The power supply unit is a critical part of this irrigation system as it powers the STM32 microcontroller, GSM module, relays, pump, and sensors. By delivering the correct voltage to each module, it ensures smooth operation of the hardware and prevents damage to sensitive electronic components. It also provides stable regulation to avoid voltage fluctuations that could disrupt the system. Additionally, it supports battery or solar integration, making the setup reliable even in remote locations.

2.2.8 Wires, Connectors, Breadboard/PCB

Overview

Wires, connectors, and circuit boards form the essential backbone for interconnecting all electronic components in the system. They ensure proper electrical connections, signal transmission, and mechanical stability of the hardware setup.

Working Principle

Wires carry power and signals between different modules such as the microcontroller, GSM module, relays, pump, and sensors. Connectors allow easy plugging and unplugging of components, improving modularity and ease of maintenance. During prototyping, a breadboard is often used to test circuits without soldering, while for the final implementation, a PCB (Printed Circuit Board) is designed to permanently mount and interconnect the components in a neat and reliable manner. Proper wire management helps prevent loose connections, noise interference, and short circuits. Heat-shrink tubing and cable sleeves protect joints from environmental damage. Color-coded or labeled wires make troubleshooting easier, ensuring long-term reliability and maintenance-friendly operation.

Technical Specifications

• Heat-shrink tubing used for safe insulation and protection of joints
• Jumper wires for easy testing and temporary circuit setups
• Cable ties for neat wire management and secure placement
• Screw terminals for firm and reliable power line connections
• Appropriate wire gauges selected based on current requirements of each module

Advantages

- Provide reliable connections between modules
- Allow flexibility during prototyping and testing
- Ensure durability and long-term stability in the final design
- Make assembly, debugging, and maintenance easier

Role in Project

In this irrigation system, wires and connectors are used to establish all necessary connections between the STM32 controller, GSM module, sensors, relays, pump, and solenoid valves. A breadboard may be used in the testing phase, while a custom PCB can be employed for the final build to achieve a more compact, professional, and durable design. breadboards are not reliable for long-term use in agricultural environments. Protective accessories like heat-shrink tubing improve insulation and safeguard joints from moisture and dust. Organized and color-coded wiring simplifies debugging and future maintenance, ensuring long-term reliability of the irrigation system.

2.2.9 Soil Moisture Sensor

Fig 2.8 Soil Moisture

Overview

A soil moisture sensor is a sensing device used to measure the amount of water present in the soil. It helps determine whether irrigation is required, making it an essential part of smart farming systems. By providing real-time feedback on soil conditions, it ensures that crops receive just the right amount of water.

Working Principle

The soil moisture sensor works by measuring the electrical resistance or capacitance of the soil. When the soil is dry, its resistance is higher, while moist soil conducts electricity more easily. The sensor converts this property into an electrical signal that is read by the microcontroller. Based on this input, the controller decides whether to activate or stop irrigation.

Technical Specifications

- Provides analog or digital output depending on the type
- Sensitive to soil water content

- Easily interfaced with microcontrollers
- Suitable for continuous or interval-based monitoring

Advantages

- Ensures precise and need-based irrigation
- Reduces water wastage by preventing unnecessary watering
- Helps maintain healthy soil conditions for crop growth
- Low cost and easy to install in fields

Role in Project

In this irrigation system, the soil moisture sensor continuously monitors the water content of each field. When the moisture level drops below a set threshold, the STM32 microcontroller activates the water pump and solenoid valves to irrigate the field. Once the required moisture level is reached, the system automatically turns off irrigation and sends an SMS update to the farmer. This sensor plays a vital role in optimizing water usage and improving crop yield.

2.2.10 Dry Run Sensor

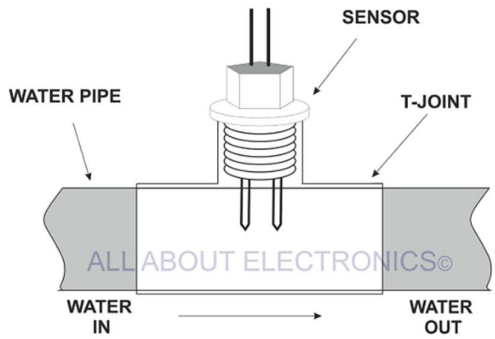

Fig 2.9 Dry Run Sensor

Overview

A dry run sensor is a protective device used to detect the absence of water flow in pumping systems. It prevents the pump from operating when there is no water, which could otherwise lead to overheating, mechanical wear, or permanent motor damage.

Working Principle

The dry run sensor works by monitoring either the water flow or the electrical characteristics of the pump (such as current consumption). If no water is detected while the pump is running, the sensor signals the microcontroller, which immediately turns off the pump. This ensures that the motor is only active when water is available, protecting it from dry running conditions.

Technical Specifications

- Detects absence of water flow or reduced pressure
- Provides digital or analog signal output to the controller
- Compatible with microcontroller-based systems
- Can be installed in line with the pump or in the supply tank

Advantages

- Protects the pump from damage due to dry running
- Reduces maintenance costs and extends pump lifespan
- Provides automatic and reliable motor protection
- Simple integration with microcontrollers and automation systems

Role in Project:

In this irrigation system, the dry run sensor ensures that the water pump does not operate if there is no water in the source. The STM32 microcontroller continuously monitors the sensor's output, and if a dry run condition is detected, the system stops the pump immediately and sends an SMS alert to the farmer. This safety feature makes the irrigation system more reliable and prevents costly damage to the equipment.

2.2.11 ESP 32:

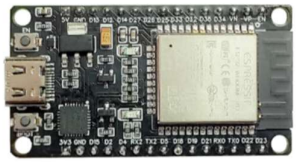

Fig 2.10 ESP-32 Microcontroller

The ESP32 is a low-cost microcontroller with built-in Wi-Fi and Bluetooth, making it ideal for IoT-based applications like smart irrigation. It collects sensor data and either processes it locally to control actuators (pump/valve) or sends it to a smartphone/cloud via Wi-Fi or Bluetooth for remote monitoring and control.

Role in Project:

Provides remote monitoring and control, allowing users to check soil moisture levels and operate irrigation devices from anywhere using Wi-Fi or Bluetooth.

Advantages:

- Integrated wireless connectivity.
- Low power and cost effective.
- Multiple ADC, DAC, PWM and GPIO pins for sensor interfacing
- Fast response for real-time applications

2.3 Software Requirements:

1. STM32CubeIDE: Firmware development and programming STM32F103C8T6.
2. Embedded C (HAL): Used for coding in STM32.
3. USART (HAL): To receive commands from GSM module.
4. I²C/SPI: If you use I²C LCD for fewer GPIO pins.
5. MIT App Inventor: Designing Android App with buttons to send predefined SMS

The software requirements for this project focus on programming, communication, and application development. STM32CubeIDE is used for firmware development and programming the STM32F103C8T6 microcontroller, providing an efficient environment for compiling and debugging embedded applications. The system is programmed using Embedded C with Hardware Abstraction Layer (HAL) libraries, which simplify peripheral handling and ensure reliable sensor and actuator control. For communication with the GSM module, the USART (HAL) interface is utilized to send and receive SMS-based commands, enabling smooth interaction between the controller and the farmer. Additionally, I2C or SPI communication can be employed when interfacing with an I2C-enabled LCD, reducing the number of GPIO pins required on the microcontroller. On the user side, an Android application is designed using MIT App Inventor, which provides a simple graphical interface with predefined buttons to send SMS commands directly to the irrigation system.

2.3.1. STM32CubeIDE

Overview

STM32CubeIDE is the official Integrated Development Environment (IDE) from STMicroelectronics, specifically designed for STM32 microcontrollers. It combines project management, code editing, and peripheral configuration, compiling, and debugging tools into one software package. It integrates STM32CubeMX, a graphical configuration tool, which makes setting up hardware peripherals (like GPIO, ADC, UART, timers, I²C, and SPI) more efficient. STM32CubeIDE is widely used in IoT, robotics, and embedded automation projects because of its powerful debugging support and low-level hardware access.

Key Features

1. **Code Editor** – Provides a user-friendly interface with syntax highlighting, auto-completion, and error detection for Embedded C/C++.
2. **Peripheral Configuration** – Built-in CubeMX tool to graphically configure peripherals and automatically generate initialization code.
3. **Debugger Support** – Includes GDB (GNU Debugger) with features like breakpoints, memory watch, and variable inspection for error tracking.
4. **Library Integration** – Supports CMSIS, HAL, and LL libraries, making code portable across STM32 families.
5. **Cross-Platform** – Compatible with Windows, Linux, and macOS.

Usage in project:

In this irrigation system, STM32CubeIDE is used to develop the firmware for the STM32F103C8T6 microcontroller. The firmware handles soil moisture sensing, pump and solenoid valve controls communication, dry run protection, and LCD display updates. For example, Cube MX is used to configure the UART interface for SIM800L communication and the ADC for soil moisture sensor input.

2.3.2 Embedded C with HAL Libraries

Overview

Embedded C is the programming language used to control microcontrollers in real-time systems. With STM32, the Hardware Abstraction Layer (HAL) simplifies programming by providing pre-built functions to control peripherals, eliminating the need for complex register-level coding. HAL makes the code more readable, portable, and less error-prone.

Key Features

1. **Abstraction Layer** – Functions like HAL_GPIO_WritePin () or HAL_UART_Transmit () simplify hardware access.
2. **Peripheral Drivers** – Includes libraries for GPIO, ADC, timers, communication protocols (UART, I²C, SPI), and more.
3. **Portability** – Code can be reused across different STM32 microcontrollers with minimal changes.
4. **Real-Time Efficiency** – Ensures fast and predictable responses for time-critical tasks.

Usage in Project

In this irrigation project, Embedded C with HAL is used to write the irrigation logic. The ADC

(Analog-to-Digital Converter) is programmed using HAL to measure soil moisture levels.Timer peripherals are configured using HAL for irrigation duration management. GPIO pins are controlled using HAL functions to switch relays for the pump and solenoid valves.

3. UART (HAL)

Overview

USART (Universal Synchronous/Asynchronous Receiver/Transmitter) is a serial communication protocol that allows data transfer between the STM32 microcontroller and external modules. In this system, USART is crucial for GSM communication using AT commands. HAL libraries make the configuration and use of USART simpler by providing high-level functions.

Key Features

1. **Bidirectional Communication** – Allows sending and receiving data simultaneously.
2. **AT Command Compatibility** – Works seamlessly with GSM modules like SIM900A.
3. **Interrupt and DMA Modes** – Supports efficient communication for real-time applications.
4. **Debugging Support** – Enables serial communication monitoring using tools like PuTTY or Arduino Serial Monitor.

Usage in Project

USART is used by the STM32 to communicate with the SIM800L GSM module. For example:

- To send an SMS, the STM32 issues the AT command: AT+CMGS="phone number".
- The GSM module replies with an acknowledgment, and then the STM32 transmits the irrigation status message.
- Received SMS messages from the farmer (e.g., “Field 1, 10 minutes”) are processed by the STM32 through USART.

This makes SMS-based control possible even in rural areas without internet access.

4. I²C/SPI

Overview

I²C (Inter-Integrated Circuit) and SPI (Serial Peripheral Interface) are communication protocols used to connect microcontrollers with peripherals such as displays, sensors, or memory devices. Both protocols are faster and more efficient compared to standard parallel communication. I²C reduces pin usage, while SPI provides higher data rates.

Key Features

1. **I²C Protocol** – Uses only two wires (SDA and SCL), supports multiple devices on the same bus.
2. **SPI Protocol** – Uses four wires (MISO, MOSI, SCK, SS), provides high-speed full-duplex communication.
3. **Flexibility** – Compatible with various devices such as LCDs, RTCs, and external EEPROMs.
4. **Scalability** – I²C allows multiple devices to be connected with unique addresses.

Usage in project

In this irrigation system, I²C/SPI can be used for interfacing the 16x2 LCD display. Instead of using multiple GPIO pins for parallel communication, an I²C/SPI-based LCD reduces pin usage, leaving more GPIOs available for sensors and relays. This makes the design more compact and efficient.

5. MIT App Inventor

Overview

MIT App Inventor is a visual, block-based programming platform used to create Android applications easily without traditional coding. Its simplicity makes it ideal for designing IoT control interfaces that communicate with hardware systems.

Key Features

1. **Drag-and-Drop Designer** – Simple interface for designing app layouts with buttons, text boxes, and labels.
2. **Block-Based Coding** – Logical flow built by arranging code blocks instead of writing traditional code.
3. **SMS Component** – Built-in tools for sending and receiving SMS messages directly from the app.
4. **Cross-Platform Deployment** – Apps can run on any Android smartphone.

Usage in project

The Android app developed using MIT App Inventor provides the farmer with a user-friendly control panel. Predefined buttons allow the farmer to send SMS commands such as *"*Start Pump*"* or *"*Field 2 – 15 minutes*"* without manually typing them. The app also receives SMS updates from the system, displaying status messages like *"*Irrigation completed successfully*"* or *"*Dry run detected*"*. This makes the system more accessible to farmers with limited technical skills.

CHAPTER 3

IMPLIMENTATION

The system is implemented by interfacing soil moisture, pressure, and flow sensors with the STM32 controller to monitor field conditions. The ESP32 module connects the system to the MIT App for wireless monitoring and control. GSM communication enables SMS-based operation in remote areas without Wi-Fi. Automated timer, dry-run protection, and field-wise control ensure efficient and reliable irrigation management.

3.1 Methodology

The methodology of the proposed IoT-Controlled Automated Multi-Field Irrigation System involves a structured approach covering design, development, and implementation stages. The process began with requirement analysis, where the need for an automated and remotely controlled irrigation solution was identified. The main objectives were to minimize water wastage, reduce manual effort, and safeguard the pump from dry run conditions. Essential farmer-centric features such as SMS-based control, multi-field operation, and real-time feedback were established as key requirements to ensure practical usability and efficiency in agricultural environments.

In the hardware and software design phase, components like the STM32F103C8T6 microcontroller, SIM800L GSM module, soil moisture and dry run sensors, relay modules, solenoid valves, and an LCD display were selected and integrated to form the physical system. The STM32 acts as the central unit, processing sensor data and controlling irrigation components. The firmware was developed using STM32CubeIDE in Embedded C with HAL libraries to ensure robust performance. UART communication was configured for GSM message handling, while ADC inputs were used to acquire sensor data. Additionally, an Android-based application was created using MIT App Inventor, allowing farmers to send predefined SMS commands for pump and field control through a simple user interface. Finally, the system integration, testing, and implementation stages ensured the seamless interaction between hardware and software. The GSM module communicated with the STM32 via USART, receiving SMS commands and triggering the appropriate irrigation responses based on soil conditions and timer settings. The LCD displayed real-time system status, and feedback messages were sent to the user confirming actions. Extensive testing validated the performance of pump protection, moisture-based automation, and timer-controlled irrigation. The completed system was successfully implemented as a modular, scalable, and reliable solution capable of managing multiple fields efficiently while optimizing water use and ensuring the safety of irrigation equipment.

3.2 Circuit Diagram:

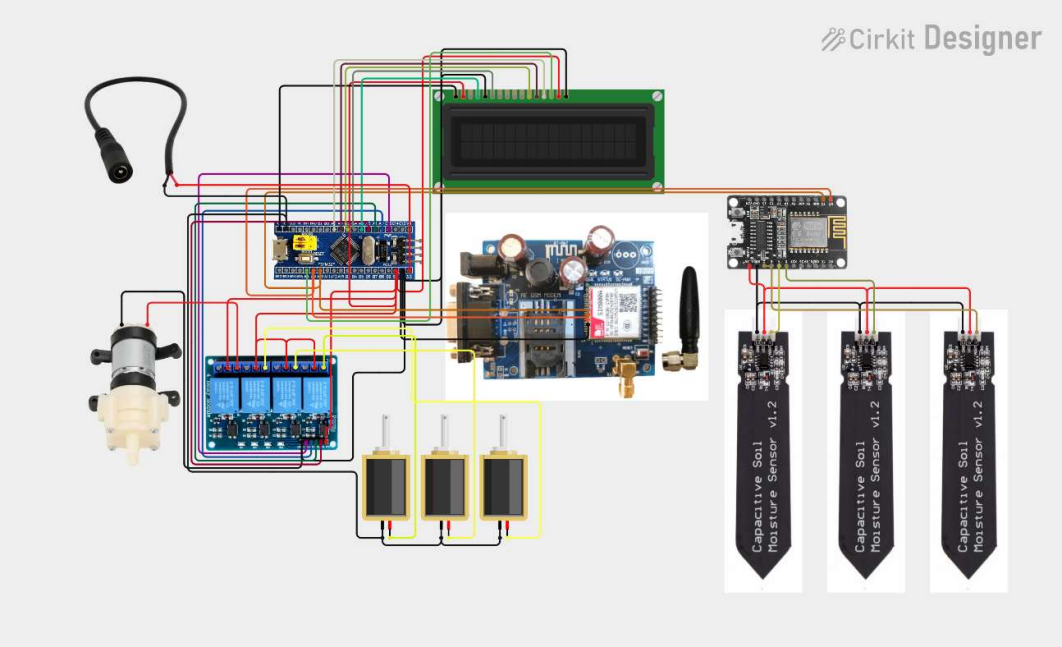

Fig 3.1 Circuit Diagram for Automated Irrigation System Operation

The system uses three capacitive soil moisture sensors to continuously monitor the moisture levels in different fields, sending accurate analog signals to the STM32 microcontroller. The STM32 processes these readings through its ADC channels and compares them with predefined threshold values to determine whether irrigation is required. When the soil becomes dry, the microcontroller activates the relay module through its GPIO pins, which in turn switches the water pump and the corresponding solenoid valves. The water pump draws water from the main storage tank, while the solenoid valves ensure that water is supplied only to the field where moisture is low. For remote control, the GSM module plays a key role by receiving commands through SMS, such as “ALL VALVES ON” or “PUMP OFF,” enabling the user to operate the system from any location. In addition to SMS control, the ESP8266 Wi-Fi module uploads real-time soil moisture data to the cloud, enabling IoT-based monitoring and providing graphical insights through online dashboards. The entire setup is powered by a reliable power supply adapter that delivers regulated voltages to all the components including the STM32, GSM module, ESP8266, LCD display, sensors, and relay drivers. This ensures stable operation, protects components from voltage fluctuations, and supports uninterrupted irrigation control. The integration of both GSM and cloud connectivity ensures dual-mode monitoring, offering flexibility for users with or without internet access. Its automated control logic minimizes manual intervention, reducing labor while improving irrigation accuracy. Overall, the system provides a robust and scalable solution for modern agricultural automation.

3.3 Flow of the project

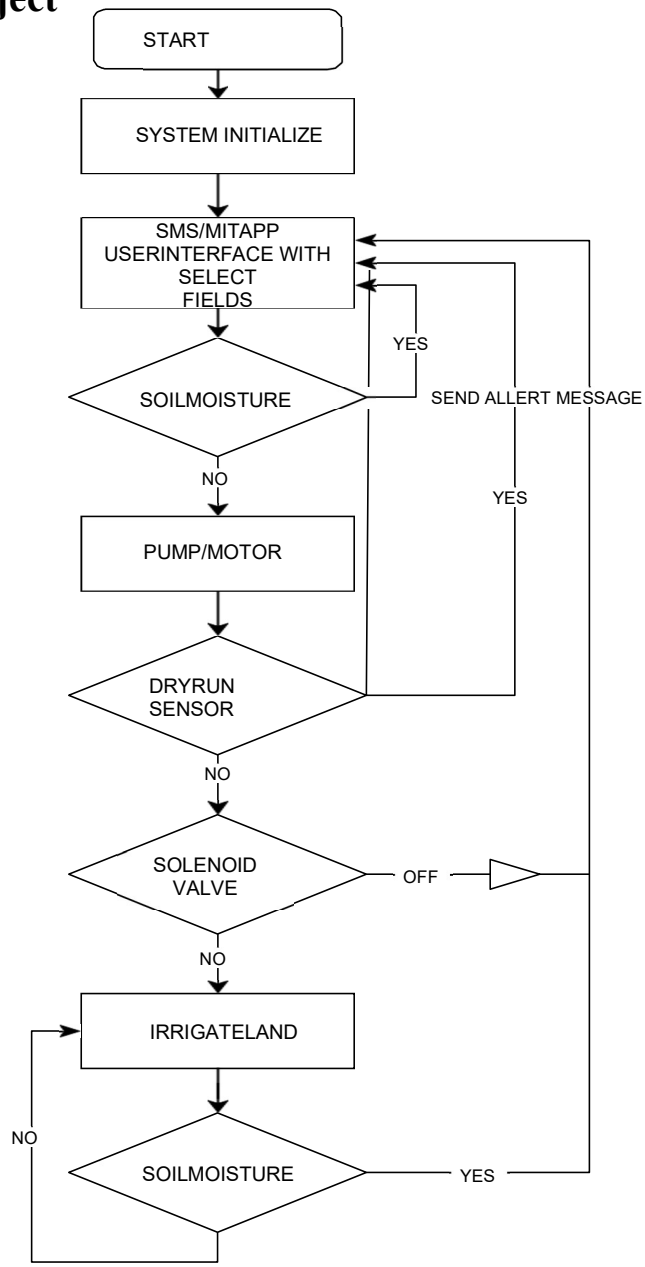

Fig 3.2 Flowchart for Automated Irrigation System Operation

The flowchart of the Automated Irrigation System shows the sequence of decisions and actions that the system follows during operation. The process begins with system initialization, where the STM32 microcontroller activates all sensors and prepares the GSM module for communication. The farmer can then send an SMS command or use the mobile application to select a particular field and set the irrigation duration. Once a command is received, the first step is to check the dry run sensor to ensure that water is available. If the sensor detects that there is no water flow, the system immediately halts irrigation and sends an SMS alert to the farmer (e.g., *"*Irrigation failed: No water supply detected*"*), thus protecting the pump from damage.

If water is available, the system next verifies the soil moisture level of the selected field. If the soil is already moist enough, irrigation is skipped, and the farmer is notified (e.g., *"Field 1 already has sufficient moisture, irrigation not required"*). On the other hand, if the soil is dry, the system proceeds to activate the pump via the motor driver and opens the solenoid valve corresponding to the chosen field.

The irrigation continues until one of two conditions is met:

1. The preset timer duration (set by the farmer) runs out.
2. The soil moisture sensor detects that the required moisture threshold has been reached before the timer ends.

At that point, the pump and solenoid valve are turned off automatically, and a confirmation message is sent to the farmer (e.g., *"Irrigation for Field 2 completed successfully"*).

Example Scenario: Suppose a farmer sends the SMS *"Field 3, 20 minutes"*. The system initializes and checks the dry run sensor. If water is available, it checks the soil moisture in Field 3. Let's assume the soil is dry. The system then switches on the pump and opens the solenoid valve for Field 3. Irrigation begins, and water flows to the field. After 12 minutes, the soil moisture sensor detects that the soil has already reached the optimal moisture level. In this case, the system automatically stops irrigation before the 20-minute timer completes, saving 8 minutes' worth of water. The farmer then receives an SMS stating *"Field 3 irrigation stopped after 12 minutes – moisture level adequate"*.

This smart decision-making process not only prevents water wastage and reduces manual labor but also ensures that the crops always receive the right amount of water at the right time.

3.4 Algorithm for Automated Irrigation System

1. **Start the System**
    - Initialize STM32 microcontroller, GSM module, sensors, and peripherals.
2. **Receive Command**
    - Farmer sends SMS or selects a field using the Android app.
    - System receives command via GSM module (through USART).
3. **Check Dry Run Condition**
    - Read the dry run sensor to verify water availability.
    - If water is not available:
        - Stop the process.

        - Send alert SMS to farmer (*"Dry run detected, irrigation stopped"*).
        - Go back to idle state.

4. **Check Soil Moisture Level**
    - Read soil moisture sensor value of the selected field.
    - If moisture is sufficient:
        - Skip irrigation.
        - Send SMS update ("Soil already moist, irrigation not required").
        - Return to idle state.
5. **Start Irrigation**
    - Activate the pump using the relay module.
    - Open the solenoid valve of the selected field.
6. **Monitor Irrigation Process**
    - Continue irrigation until one of the conditions is met:
        - The **preset timer duration** expires.
        - The **soil moisture sensor** detects adequate moisture.
7. **Stop Irrigation**
    - Turn OFF the pump and close the solenoid valve.
    - Send SMS confirmation to farmer ("Irrigation completed successfully").
8. **End/Wait for Next Command**
    - System returns to idle state, ready for the next SMS or field selection.

3.5 Test Cases:

Test cases are a critical component of software testing. They are specific scenarios or situations that are designed to verify whether a particular feature, function, or system behaves as expected. Test cases outline the steps to be executed, the input data to be used, and the expected outcomes. The main purpose of test cases is to ensure that the software or system meets the specified requirements and functions correctly in different scenarios by systematically executing test cases, you can identify defects, errors, or unexpected behaviors and address them before the software is deployed to production or released to end- users.

3.5.1 HARDWARE IMPLEMENTATION:

Activation: When the system is powered on, it goes through an initialization process to ensure that all key components are ready and functional as shown on figure. This is crucial because the system relies on multiple hardware components working together in real time. Here's a breakdown of each component's role in this process:

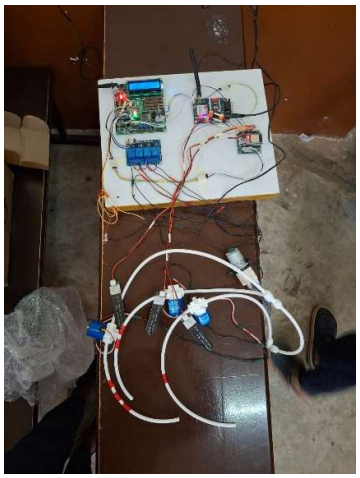

Fig 3.3 System Initialization and Processing

1. STM32F103C8T6 Microcontroller (Blue Pill)

Role: Acts as the brain of the irrigation system, managing sensor inputs, controlling actuators, and coordinating GSM-based communication. It continuously monitors environmental parameters such as soil moisture and executes automated irrigation decisions. The microcontroller also manages timing functions, dry-run protection, and display updates.

Initialization: Before execution, GPIO pins are configured for digital input/output operations, ADC channels for analog sensor readings, and UART for GSM communication. I²C or SPI interfaces are initialized for peripherals like the LCD or sensors if needed. Timers, interrupts, and clock settings are configured to ensure accurate and reliable performance.

2. SIM900A GSM Module

Role: Facilitates remote monitoring and control by enabling bidirectional communication between the user and the irrigation system. Through SMS alerts, it can send real-time updates like soil status or pump activity, and receive control commands from the farmer to start or stop irrigation remotely. **Initialization:** Connected via UART to the STM32, it requires setting an appropriate baud rate for stable communication. The module's readiness is confirmed through AT commands such as network registration and signal quality checks. A test message is transmitted to verify successful communication before system integration.

3. Relay Module (4-Channel)

Role: Acts as an interface between the low-power microcontroller and high-power devices like pumps and solenoid valves. Each relay channel can independently control an external circuit, enabling selective irrigation of multiple fields or zones.

Initialization: The control pins are configured as digital outputs on the microcontroller. Each relay is tested by toggling ON/OFF signals and observing the corresponding device response. Proper isolation and flyback diodes are verified to prevent damage from back-EMF generated by inductive loads.

4. Water Pump (12V DC)

Role: Pumps water to the irrigation lines when triggered by the controller. Provides water pressure for irrigation lines when activated by the system. It forms the primary actuator responsible for delivering water from the source to the desired field area.

Initialization: The pump is connected to the relay output with adequate power supply capacity. Compatibility of current and voltage ratings is verified before operation. A short-duration test run is conducted to confirm smooth operation, and dry-run protection is tested to prevent damage under no-water conditions.

5. Solenoid Valves (3 Nos)

Role: Controls water flow to specific fields or zones under microcontroller commands. Control the distribution of water to different zones or fields. The system selectively opens or closes these valves based on moisture data or user commands, ensuring optimized water usage and preventing over-irrigation.

Initialization: Each valve is assigned a control pin linked to a relay channel. Functionality is verified by toggling ON/OFF signals to test open and close operations. The timing and responsiveness of valve actuation are checked to ensure proper field division management.

6.16x2 LCD Display

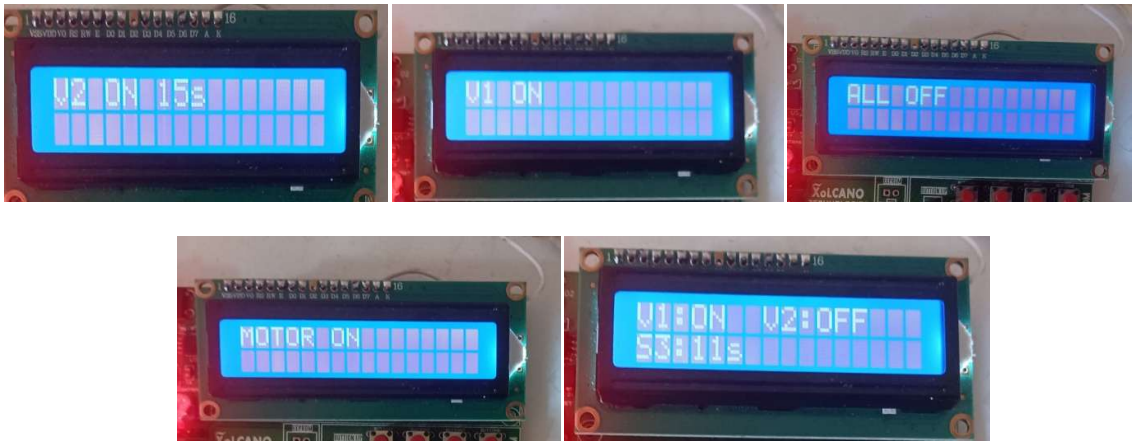

Fig 3.4 Display Status of Motor and Valves on 16x2 LCD Display

Role: Displays live system data such as soil moisture readings, pump status, and GSM communication messages. Serves as a user interface displaying key system parameters such as soil moisture values, water pump status, GSM signal strength, and received messages. It enhances user interaction and system transparency.

Initialization: Configured in either 4-bit or 8-bit mode depending on available pins. After powering up, the display is cleared, and cursor positioning is verified. Test strings or data values are printed to ensure accurate character rendering and stable operation.

7. Power Supply Unit

Role: Provides stable DC power to all system components. Provides stable and regulated DC voltage to all modules, ensuring consistent and safe operation. It isolates and protects sensitive electronics from voltage fluctuations and power surges.

Initialization: Voltage levels (e.g., 3.3V and 5V rails) are verified using a multimeter before connecting components. The current supply capacity is checked to handle peak loads, particularly for the pump and GSM module. Voltage regulators and capacitors are tested to confirm ripple-free output.

8. ESP32 (Microcontroller)

Role: The ESP32 acts as the main controller that reads soil-moisture data and decides when irrigation is needed. It controls the pump and solenoid valves through the relay module for automatic field-wise watering. It also handles SMS communication for remote commands and system status updates.

Initialization: During startup, the ESP32 configures sensors, relay pins, and the GSM module for proper operation. It loads moisture thresholds and system parameters required for decision-making. After successful initialization, the controller becomes ready to execute the irrigation process.9. Soil Moisture Sensor

9.Soil Moisture

Role: Measures the volumetric water content in the soil and sends corresponding analog voltage signals to the microcontroller. This data helps the system determine when to activate or deactivate irrigation for optimal soil moisture levels.

Initialization: Connected to one of the STM32's ADC pins, the sensor output is calibrated using dry and saturated soil samples. The analog voltage response is monitored and mapped to percentage moisture levels for accurate decision-making.

10. Dry Run Sensor

Role: Detects whether water is flowing through the pump to prevent it from running dry, which can cause mechanical damage. It ensures system safety and prolongs pump life by automatically stopping operation in dry conditions.

Initialization: The sensor output is connected to a digital or analog input pin on the microcontroller. Testing involves simulating both water-present and no-water conditions to verify correct detection. Once validated, it is integrated into the control logic for automated pump shutdown.

3.5.2 SOFTWARE IMPLEMENTATION:

SMS Command:

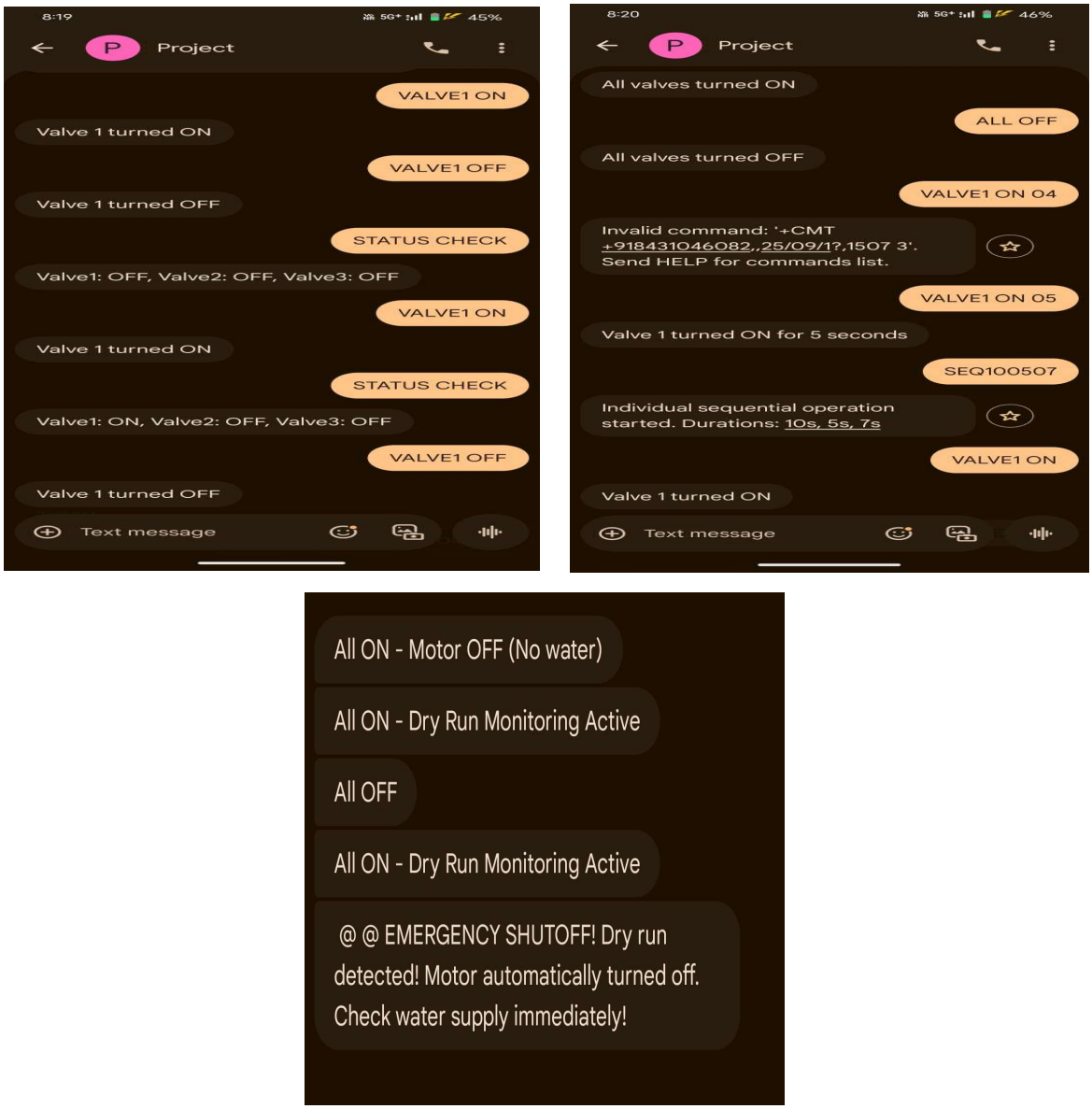

Fig 3.5 Snapshot of SMS Commands

- **Message Communication**

- The GSM module receives an SMS sent from the user's mobile phone.
- The message is read by the microcontroller (like Arduino, Raspberry Pi, or similar), which checks what command is sent.
- Based on the content of the SMS, the system performs the required operation.

- **Valve Control**

- Each valve (for example, Valve1, Valve2, and Valve3) is connected to a relay that acts as a switch.
- When the controller receives the command VALVE1 ON, it activates the relay connected to Valve1, allowing water (or air, or other fluid) to flow.
- When VALVE1 OFF is received, it deactivates that relay, stopping the flow.

- **Command Execution and Response**

- After executing each command, the system sends a confirmation message back to the user, confirming the action taken.
- For example, when you send VALVE1 ON, you get a reply saying "Valve 1 turned ON", and when you send VALVE1 OFF, it replies "Valve 1 turned OFF".
- This ensures that the user knows the exact state of the valves even from a remote location.

- **Status Check**

- When the user sends the message STATUS CHECK, the system replies with the present condition of all valves.
- For example: "Valve1: ON, Valve2: OFF, Valve3: OFF".
- This helps to monitor the system anytime without physically being near it.

- **Timed Operation**

- The project also supports timed control.
- When the user sends a command like VALVE1 ON 05, the system turns ON valve 1 for 5 seconds and then automatically turns it OFF.
- This is useful in irrigation or process control where each valve needs to operate for a fixed period.

- **Sequential Operation**

- The command SEQ100507 starts a sequence where multiple valves turn ON and OFF one after another for specific durations.
- For example, this command makes:
  - Valve 1 ON for 10 seconds
  - Valve 2 ON for 5 seconds
  - Valve 3 ON for 7 seconds
- The system then automatically executes this sequence and informs the user:
  "Individual sequential operation started. Durations: 10s, 5s, 7s."

- **Error Handling**

- If a message is sent in the wrong format, the system shows an error like:
  "Invalid command: … Send HELP for commands list."
- This helps the user understand that the command was not recognized and should be corrected.

Example

- You sent → VALVE1 ON
  → The system replied → "Valve 1 turned ON"
- You sent → VALVE1 OFF
  → The system replied → "Valve 1 turned OFF"
- You sent → MOTOR ON
  → The system replied → "Motor turned ON
- You sent → MOTOR OFF
  → The system replied → "Motor turned OFF"
- You sent → STATUS CHECK
  → The system replied → "Valve1: ON, Valve2: OFF, Valve3: OFF"
- You sent → VALVE1 ON 05
  → The system replied → "Valve 1 turned ON for 5 seconds"
- You sent → SEQ100507
  → The system replied → "Individual sequential operation started. Durations: 10s, 5s, 7s"
- You sent → INVALID COMMAND
  → The system replied → "Invalid command! Please enter a valid instruction."

MIT SOFTWARE

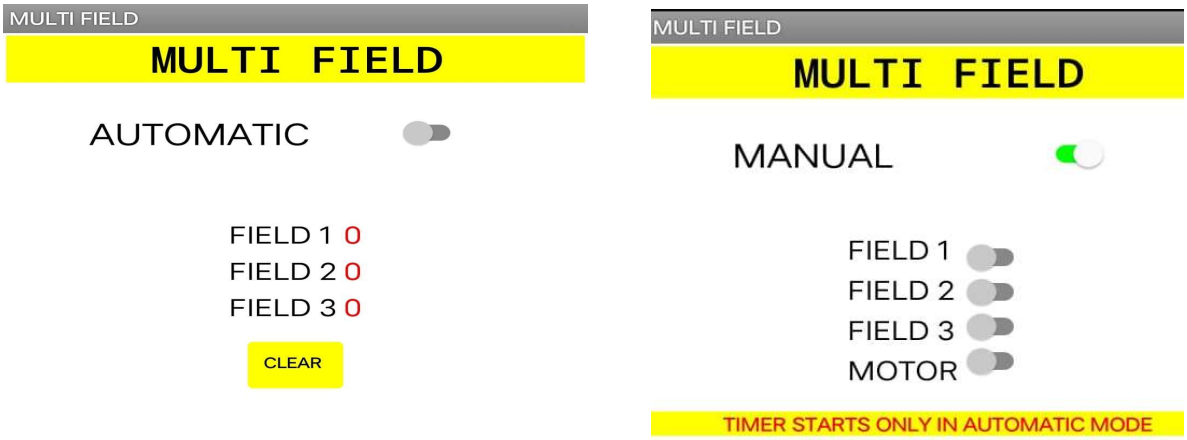

Fig 3.6 Snapshot of MIT App

The Android application interface developed using MIT App Inventor for the IoT-Enabled Multi-Field Smart Irrigation System.

This app provides a simple, user-friendly platform for farmers to control irrigation across multiple fields using SMS commands.

- The title bar displays "MULTI FIELD", indicating that the system supports multiple field zones.
- The "AUTOMATIC" toggle switch allows the user to switch between manual control and automatic mode:
  - When OFF, the farmer can manually start or stop irrigation for each field.
  - When ON, the system automatically controls irrigation based on soil moisture levels and timer settings.
- Below the toggle, the screen lists:
  - FIELD 1, FIELD 2, FIELD 3, each followed by a status indicator (0 or 1).
  - A value of 0 (in red) means the field's irrigation valve is currently OFF, while 1 (in green) would indicate that the valve is ON and watering that field.

This clean and minimal interface ensures that farmers — even with little technical experience — can easily monitor and control irrigation for multiple fields from their mobile device. The design emphasizes simplicity, real-time status visibility, and ease of use in rural environments.

MIT App Inventor – Implementation Details

The MIT App Inventor was used to design and program the Android application that communicates with the STM32-based irrigation system via the SIM800L GSM module.

Key Features:

1. **SMS-Based Control:**
   - The app uses MIT App Inventor's "Texting" component to send predefined SMS commands such as:
     - FIELD 1 ON
     - FIELD 2 OFF
     - STATUS CHECK
   - These commands are received by the GSM module, which passes them to the STM32 controller.
2. **Automatic/Manual Toggle:**
   - The switch control enables or disables automatic irrigation mode.
   - In automatic mode, irrigation decisions are based on soil moisture data and timers.
   - In manual mode, farmers can manually select and control each field's valve.
3. **Real-Time Status Display:**
   - The app receives SMS updates from the GSM module (e.g., "Field 1 irrigation completed").
   - The status values (0 or 1) next to each field update automatically to show the current irrigation state.
4. **Error Notifications:**
   - If a command is incorrect or the GSM connection fails, the app displays error alerts like "Invalid command" or "Network not available."
5. **User Interface Design:**
   - Created using drag-and-drop blocks in MIT App Inventor.
   - Each field and toggle are associated with SMS sending blocks.
   - Bright colors (yellow header, red status) are used for better visibility under outdoor light conditions.

CHAPTER 4

EXPERIMENT ADVANTAGES AND APPLICATIONS

4.1 Advantages

- **Works With or Without Internet:**
  The system operates using SMS, making it ideal for rural areas with poor or no internet connectivity.
- **Manages Multiple Fields Separately:**
  Farmers can control irrigation in different fields independently, based on the specific needs of each crop.
- **Real-Time Alerts via SMS:**
  Farmers receive instant notifications about system activity, water levels, and faults, improving responsiveness.
- **Automatic Dry Run Protection:**
  Prevents pump damage by detecting the absence of water flow and automatically shutting off the motor.
- **Saves Time and Reduces Physical Labor:**
  Eliminates the need for farmers to manually visit fields to start or stop irrigation, saving time and effort.
- **Scheduled Irrigation with Timer Control:**
  Timer-based operation ensures water is supplied for the correct duration, avoiding overwatering.
- **Efficient Water Usage:**
  Irrigates only when needed, based on sensor data, significantly reducing water wastage.
- **Minimizes Human Errors:**
  Automation ensures that irrigation is done accurately, reducing mistakes caused by manual operation.
- **Improves Crop Yield and Health:**
  Consistent and timely watering leads to better plant growth and higher productivity.
- **User-Friendly Operation:**
  Simple SMS commands make the system easy to use, even for farmers with limited technical knowledge.

- **Supports Sustainable Agriculture:**
  Promotes responsible water management practices, contributing to long-term environmental sustainability.

4.2 Applications:

- Multi-Field Farm Management: Ideal for large farms with multiple fields requiring different irrigation schedules and control.
- Remote Operation via SMS: Farmers can control and monitor irrigation systems remotely using simple SMS commands, eliminating the need for internet access.
- Automated Dry Run Protection: Prevents pump damage by automatically turning off the motor when water supply is insufficient.
- Timer-Based Irrigation: Enables scheduled watering, optimizing water usage and ensuring consistent crop hydration.
- Energy and Water Efficiency: Reduces water waste and energy consumption through intelligent automation and monitoring.
- Reduction: Minimizes the need for constant human supervision, saving time and labor costs.
- Suitable for Rural Areas: Works well in areas with poor internet connectivity but reliable GSM networks.

CONCLUSION AND FUTURE SCOPE

Conclusion

The proposed IoT-enabled smart irrigation system successfully automates the process of water management for multi-field farming using GSM-based communication. By integrating the STM32 microcontroller with sensors such as soil moisture and dry-run detectors, the system ensures that irrigation occurs only, when necessary, thus conserving both water and energy. The timer-based control and automatic dry-run protection features enhance system reliability and prevent equipment damage. Through the MIT App Inventor-based Android application, farmers can remotely control and monitor irrigation operations using simple SMS commands, even in regions with poor internet connectivity. This eliminates the need for continuous field visits and manual supervision. The project demonstrates how low-cost, energy-efficient automation can significantly improve agricultural productivity, promote sustainable water use, and reduce human effort. Overall, the system provides a practical and farmer-friendly solution that bridges the gap between traditional irrigation and modern smart farming practices.

Future Scope

1. IoT Cloud Integration:
   Future versions can be linked to cloud platforms like IBM Cloud or Thing Speak to log data for real-time monitoring, analytics, and long-term crop performance evaluation.
2. Weather-Based Irrigation:
   Integration of weather APIs or local sensors could enable predictive irrigation, automatically adjusting water schedules based on rainfall forecasts and temperature conditions.
3. Solar Power Implementation:
   Using solar panels to power the system will make it more sustainable and independent from the electrical grid, ideal for rural or off-grid areas.
4. Machine Learning and AI:
   Incorporating AI models can help analyse soil, crop type, and weather data to recommend the optimal irrigation duration and frequency for each field.
5. Mobile App Enhancement:
   The MIT App can be upgraded to include graphical dashboards, voice commands, multilingual support, and GPS-based field mapping for easier control.
6. Scalability and Multi-Sensor Networks:
   Expansion to more fields and sensor types—such as pH, water level, and nutrient sensors—can further improve precision agriculture.

Reference

[1].An Internet of Things (IoT) based Sustainable Water Management: Sreekanth Narendran, Preeja Pradeep, Maneesha Vinodini Ramesh IEEE, pp 88-96, 2017.

[2]. Internet of Things (IoT) Enabled Water Monitoring System: Thinagaran Perumal1, 1 Md Nasir Sulaiman, 2 Leong.C.Y 1 Universiti Putra Malaysia, 2 Catrino, Malaysia,IEEE, pp 86-86, 2015.

[3].AgriFusion: An Architecture for IoT and Emerging Technologies Based on a Precision Agriculture Survey:RITESH KUMAR SINGH, (Graduate Student Member, IEEE), RAFAEL BERKVENS, (Member, IEEE), AND MAARTEN WEYN, IEEE, pp 136253-136283,2021.

[4].IoT-Based Water Management Systems: Survey and Future Research Direction: SHEREENISMAIL (Graduate Student Member, IEEE), DIANA W. DAWOUD2 , (Member, IEEE), NADHEM ISMAIL, RONALD MARSH1, AND ALI S. ALSHAMI, IEEE, vol 10, pp 35942-35952,2022.

[5]. Design of Smart Farm Irrigation Monitoring System Using IoT and LoRa: Kurniawan D. Irianto Jurnal RESTI (Rekayasa Sistem dan Technology iInformasi) Vol. 6 No.1, pp 47-55, (2022)

[6]. IoT-Enabled Smart Drip Irrigation System Using ESP32: Gilroy Pereira , Mohamed Zachary, Fawwad Daroge MDPI, pp 221-243 ,2023.

[7]. GSM based Automatic Irrigation Control System for Efficient Use of Resource and Crop Planning by Using an Android Mobile: Pavithra D.S, M, S. Srinath IOSR –JMCE, pp 49-55, 2014.

[8]. Water Management in Agriculture: A Survey on Current Challenges and Technological Solutions ABDELMADJID SAAD, ABOU EL HASSAN BENYAMINA1, AND ABDOULAYE GAMATIE, IEEE, Vol 8, pp 38082-38097, 2020.

Appendix

Appendix – A: Software Tools Used

Key software tools used in the project:

- Arduino IDE – Programming ESP32, relay control, sensor integration
- Embedded C – Motor and valve control logic
- Circuit Designer – Circuit design and simulation
- Firebase Real-Time Database – Cloud storage for sensor data, remote monitoring, and control updates

Appendix – B: Code Snippets

Below code snippets represent major functional blocks (full code can be attached separately):

- ESP32 soil-moisture reading code
- SMS receiving and command-parsing code
- Motor/relay control code
- Multi-valve switching code
- Dry run detection code
- Timer-based irrigation scheduling code

Appendix – C: System Diagrams

- Following diagrams are included for better understanding of the system architecture:
- Block Diagram
- System Flowchart
- Moisture Sensing and Irrigation Flow Diagram
- Component Wiring Diagram (ESP32 + Sensors + GSM + Relay + Pump + Valves)
- Data Flow Diagram (Sensor → ESP32 → User via SMS)

Appendix – D: Hardware Components

- Major hardware components used:
- ESP32 Development Board
- GSM Module (SIM800L/SIM900)
- Capacitive Soil Moisture Sensors (Field 1, Field 2, Field 3…)
- Relay Module (4-Channel)
- Solenoid Valves for multi-field irrigation
- Power Supply Unit
- Water level / pressure sensor (for dry-run protection)
- Connecting wires & PCB/Breadboard

Appendix – E: Project Specifications

- Number of fields controlled: 2 / 3 / 4
- Modes: Automatic mode / SMS manual mode
- Communication: GSM SMS + serial monitoring
- Safety features: Dry-run protection, overload protection
- Control method: Relay-based switching
- Power: 5V/12V regulated supply

Appendix – F: Result Screenshots

- Include screenshots of:
- SMS commands sent by the user
- System response SMS
- Moisture reading outputs
- Pump ON/OFF logs